

# Matched-Spectrum Gates in a Primitive Möbius Tail: Exact Determinant–Divisor Normal Forms, CRT–Poisson Duality, and a Frequency-Concentration Gate

Liang Wang

School of Artificial Intelligence and Automation,  
Huazhong University of Science and Technology, Wuhan 430074, P.R. China  
wangliang.f@gmail.com

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## Abstract

TPC-18 reduces a failed symmetric fixed-shift Möbius tail block to a primitive correlation of the Ramanujan residuals  $r_R = \Lambda - \Lambda_R$  on two affine rows  $m_i j + h$ , where  $m_i = \ell_i d_i$ ,  $Q = LD$ ,  $J = X/Q$ , and the determinant is  $h(m_1 - m_2)$ . We replace that still-opaque correlation by exact divisor and spectral normal forms.

Writing

$$\Lambda_R(n) = \sum_{u|n} \lambda'_R(u),$$

we prove that the actual residual, not merely the finite model, has the identity

$$r_R(n) = - \sum_{\substack{u|n \\ u \text{ squarefree}}} \mu(u) w_R(u), \quad w_R(u) > 0,$$

with an explicit weight  $w_R$ . In the primitive sector put  $H = \text{rad } |h|$ . For a pair of target divisors  $u = ga$ ,  $v = gb$ , the integers  $g, a, b$  are pairwise coprime and squarefree, the shared sign cancels exactly,

$$\mu(ga)\mu(gb) = \mu(a)\mu(b),$$

and the simultaneous congruences are soluble precisely when  $g \mid m_1 - m_2$ . This gives an exact all-modulus determinant–divisor expansion of the primitive residual correlation.

We next calibrate the prime and finite-model conditional row densities on  $(j, H) = 1$ . Their difference is an explicit  $O_{h,\varepsilon}(L^{-1}X^\varepsilon)$  bias, whose contribution has a fixed power saving relative to the natural  $XQ$  scale. The centered finite-model covariance is

$$\sum_{u,v \leq R} \frac{\lambda'_R(u)\lambda'_R(v)}{uv} ((u,v)\mathbf{1}_{(u,v)|m_1-m_2} - 1),$$

with the natural coprimality restrictions. Every coprime  $(u, v)$  term vanishes individually. Moreover,

$$g\mathbf{1}_{g|\Delta} - 1 = \sum_{0 < r < g} \exp(2\pi i r \Delta / g),$$

so the complete-period  $j$ -zero mode is supported entirely on nonzero frequencies in the determinant  $\Delta$ .

There are then two exact analytic interfaces. For the bare dyadic, separable determinant coefficient, Fourier grouping of  $\ell_1 d_1 - \ell_2 d_2$  produces a transform that factors into two Möbius exponential sums. The full  $g, a, b$  masks and row-pair weights still require a matched reduction.

Alternatively, Poisson summation in the common  $j$ -class completes every  $H[u, v] \leq JX^{-\eta}$  term. In the remaining long-lcm range, the compatible joint residue has three coupled inverse phases of Kloosterman type; the centered single terms must be retained beside it. We prove the corresponding short/long-lcm decomposition and give the strict-margin sufficient range  $Q \leq X^{2\delta-\eta_0}$  on which the whole finite model is completable.

Finally, an exact finite witness shows that one centered finite-model row pair can have linear complete-period mass. Thus centering alone does not give rowwise cancellation in the four terms of  $r_R(n_1)r_R(n_2)$ . A separate global estimate could still use oscillation over the row variables; the matched-spectral route instead keeps the four local spectra together. Current large-sieve theorems apply to smoother or differently factorable coefficient classes and do not establish that matching for the sharp Möbius coefficient. No determinant-dispersion estimate, fixed-shift Hardy–Littlewood asymptotic, twin-prime theorem, or breach of sieve parity is claimed.

**Keywords:** Möbius tail; Ramanujan residuals; determinant spectrum; Chinese remainder theorem; Poisson summation; Kloosterman fractions; frequency concentration; sieve parity.

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## 1 Introduction

The fixed-shift prime-pair problem may be written as a sequence of normal-form and distribution gates. TPC-15 through TPC-17 isolate and assemble the factorable prefix of a symmetric Type-II packet [7–9]. TPC-18 proves that a surviving Möbius tail cannot be removed by coefficient smallness and that a failed primitive slice forces a large two-row residual correlation with distinct prime sources, large determinant, and low divisor gcd [10]. In the notation recalled below, its remaining arithmetic object has the schematic form

$$\sum_j \sum_{\alpha_1 \neq \alpha_2} b_{\alpha_1, \alpha_2}(j) r_R(m_1 j + h) r_R(m_2 j + h), \quad m_i = \ell_i d_i \asymp Q, \quad (1)$$

with determinant  $h(m_1 - m_2)$ . The point of the present paper is to replace “estimate a variable-determinant correlation” by exact arithmetic objects on which one can test a proposed dispersion input.

### 1.1 Main advances

The first advance is an exact divisor normal form for the actual residual. It is not a heuristic Ramanujan expansion. It follows from the finite divisor expansion of  $\Lambda_R$  and the classical identity  $\Lambda(n) = -\sum_{u|n} \mu(u) \log u$ . The result has a fixed sign pattern: a positive size  $w_R(u)$  multiplied by  $-\mu(u)$ . When a pair  $u, v$  is written as

$$u = ga, \quad v = gb, \quad g = (u, v),$$

the common factor  $g$  loses its Möbius sign. In a primitive affine row pair, the Chinese remainder theorem then says exactly

$$\text{joint divisibility} \iff g \mid m_1 - m_2.$$

Thus the parity-sensitive part is no longer hidden inside the values of  $r_R$ : the determinant content  $g$  is unsigned, while the transverse variables retain the signs  $\mu(a)\mu(b)$ .

The second advance is exact model centering and local prime calibration. On the allowed classes  $(j, H) = 1$ ,  $H = \text{rad } |h|$ , the prime and finite-model local densities are

$$\frac{H}{\varphi(H)} \frac{\ell d}{\varphi(\ell d)} \quad \text{and} \quad \frac{H}{\varphi(H)} \frac{d}{\varphi(d)},$$

respectively. Their difference is  $O(L^{-1}X^\varepsilon)$ , and we prove that it is negligible at the  $XQ$  obstruction scale. The remaining problem is therefore a locally calibrated spectrum-matching problem, not an unnoticed row drift; only the finite model has been proved to have exact conditional mean zero.

The third advance is a pair of dual spectral interfaces. For the finite model, the connected complete density contains

$$g\mathbf{1}_{g|\Delta} - 1, \quad \Delta = m_1 - m_2.$$

Its direct Fourier expansion produces additive phases  $\exp(2\pi i r \ell d/g)$  and a determinant-difference coefficient. Poisson summation in the common  $j$ -progression instead produces inverse phases in  $m_i$ , hence a genuine Kloosterman-type interface. These are different routes and have different coefficient obstructions.

## 1.2 A necessary negative result

It is tempting to expand

$$r_R(n_1)r_R(n_2) = \Lambda(n_1)\Lambda(n_2) - \Lambda(n_1)\Lambda_R(n_2) - \Lambda_R(n_1)\Lambda(n_2) + \Lambda_R(n_1)\Lambda_R(n_2) \quad (2)$$

and seek rowwise logarithmic saving from  $j$ -centering in all four terms. We show rigorously that this stronger strategy is false. A two-row finite example has centered finite-model covariance 433/450; repeating its complete period makes that component grow linearly. Hence centering does not erase the local  $MM$  spectrum. One may either seek additional global oscillation over the row variables or match the local spectra of all four channels before absolute values are taken. This does not disprove the residual determinant-dispersion statement of TPC-18; it rules out a stronger rowwise componentwise substitute for it.

## 1.3 Relation to current large-sieve technology

The Möbius-free smooth analogue of the direct determinant coefficient has the geometry treated by Pascadi's frequency-concentrated exceptional-spectrum large sieve: a coefficient  $\sum_{d_1\ell_1-d_2\ell_2=n} \Phi_1(d_1/D)\Phi_2(d_2/D)\exp(2\pi i d_1\alpha_1 + d_2\alpha_2)$  is covered by Theorem 1.7 of the published paper [6], subject there to the spectral-level condition  $q \gg L^2$ . That  $q$  is not the present row scale  $Q = LD$ ; no parameter identification is valid before the missing Kuznetsov reduction. The coefficient here also contains the sharp weights  $\mu(d_i)\omega_D(d_i)$ . Its Fourier transform factors exactly, but the cited theorems do not supply the required concentration estimate for those Möbius weights. Moreover, the reduction must preserve the four-channel matching in (2).

This distinction matters because the current numerical distribution frontier alone is not the missing input. Pascadi's more recent work reaches level  $5/8 - \varepsilon$  for triply well-factorable prime weights [5]; Li's version-locked bilinear theorem has the two monomial conditions used in TPC-17 [4]. Neither coefficient class contains the sharp prime–Möbius row weight left here. Classical and recent Kloosterman-fraction estimates likewise act after a suitable inverse phase has been produced [1, 3, 11]. We construct that phase algebraically, but do not assert that its three coupled moduli satisfy an existing theorem's hypotheses.

## 1.4 Scope and organization

All unconditional statements in this paper are algebraic, elementary analytic, or finite spectral identities. We do not prove cancellation of the primitive TPC-18 witness. In particular, we prove no fixed-shift Hardy–Littlewood asymptotic, no twin-prime lower bound, no new exponent of distribution, and no violation of the parity barrier.

Section 2 fixes the primitive row geometry. Section 3 derives the exact residual divisor form and the  $g, a, b$  determinant expansion. Section 4 removes the local bias. Section 5 computes

the connected finite model. [Section 6](#) gives the direct determinant Fourier gate. [Section 7](#) gives the CRT–Poisson and inverse phase gate. [Section 8](#) proves the rowwise componentwise no-go theorem. [Section 9](#) records the high-prime local tail and the exact remaining alternatives. [Section 10](#) describes the exact certificate and concludes.

## 2 The primitive row interface

Fix  $h \in \mathbb{Z} \setminus \{0\}$  and put

$$H = \text{rad } |h|. \quad (3)$$

Let  $0 < \delta < 1/2$  and

$$R = \lfloor X^{1/2-\delta} \rfloor, \quad V = \lfloor R^{1/2} \rfloor. \quad (4)$$

The finite Ramanujan model and its residual are

$$\Lambda_R(n) = \sum_{q \leq R} \frac{\mu(q)}{\varphi(q)} c_q(n), \quad r_R(n) = \Lambda(n) - \Lambda_R(n). \quad (5)$$

All target arguments below are positive integers. We use the standard convention  $c_q(n) = \sum_{a \bmod q}^* \exp(2\pi i a n / q)$ .

Let  $L, D, J, Q \geq 1$  be dyadic parameters satisfying

$$Q = LD, \quad QJ \asymp X, \quad L > \max(2R, 2|h|), \quad (6)$$

and

$$\ell \in [L/2, 2L] \text{ is prime}, \quad d \in [D, 2D] \cap [1, V]. \quad (7)$$

The integer  $d$  is squarefree, and in the primitive sector

$$(d, H) = 1, \quad (j, H) = 1. \quad (8)$$

Put

$$m = \ell d \asymp Q, \quad n_m(j) = mj + h. \quad (9)$$

Then  $(m, H) = 1$ , and (8) implies

$$(n_m(j), mH) = 1. \quad (10)$$

TPC-18 reaches row pairs  $\alpha_i = (\ell_i, d_i)$  in the generic set

$$\ell_1 \neq \ell_2, \quad |m_1 - m_2| > QX^{-\kappa}, \quad (d_1, d_2) \leq X^\kappa, \quad (11)$$

for fixed  $\kappa > 0$ . The exact identities below hold before this pruning, so any row-pair restriction independent of  $j$ , including (11), may be imposed outside them.

For a uniformly smooth compactly supported family  $F_{\alpha_1, \alpha_2}$ , supported where  $j \asymp J$  and both targets are positive, define the primitive residual correlation

$$\mathcal{C}_D(\mathcal{A}) := \sum_{(\alpha_1, \alpha_2) \in \mathcal{A}} z_{\alpha_1, \alpha_2} \sum_{\substack{j \geq 1 \\ (j, H) = 1}} r_R(n_{m_1}(j)) r_R(n_{m_2}(j)) F_{\alpha_1, \alpha_2}(j/J), \quad (12)$$

where  $|z_{\alpha_1, \alpha_2}| \ll X^\varepsilon$  absorbs the source weights, the Möbius row signs, and bounded dyadic cutoffs. Uniform smoothness means that every derivative of every  $F_{\alpha_1, \alpha_2}$  is bounded independently of  $X$  and the row pair. This hypothesis is essential in the Poisson-completion statements below.

The weighted number of individual rows is

$$\sum_{\ell \asymp L} \Lambda(\ell) \sum_{d \asymp D} \mu(d)^2 \ll LD X^\varepsilon = QX^\varepsilon. \quad (13)$$

Consequently the row-pair-orbit scale is

$$Q^2 J \asymp XQ. \quad (14)$$

This is the natural size of the TPC-18 obstruction. Every claim of negligibility below is measured against  $XQ(\log X)^{-A}$ , with a fixed power margin whenever one is asserted.

*Remark 2.1* (Smooth separation). The original TPC-18 weight contains smooth functions of  $mj/X$  and  $dj/K$ , not merely a common function of  $j/J$ . Fourier inversion in logarithmic coordinates writes such a compactly supported smooth weight as an absolutely convergent superposition of separated row and  $j$  weights. Truncating the dual variables at  $X^\varepsilon$  has arbitrary power accuracy. Thus the uniformly smooth family in (12) is the stable interface for the exact identities; no sharp  $j$ -cutoff is being silently completed.

### 3 An exact determinant–divisor normal form for the residual

We begin by converting the Ramanujan model to divisor coordinates. Set

$$\lambda'_R(u) := u \sum_{b \leq R/u} \frac{\mu(ub)\mu(b)}{\varphi(ub)} \quad (u \leq R), \quad (15)$$

and  $\lambda'_R(u) = 0$  for  $u > R$ .

**Lemma 3.1** (Finite-model divisor coefficient). *For every  $n \geq 1$ ,*

$$\Lambda_R(n) = \sum_{u|n} \lambda'_R(u). \quad (16)$$

*The coefficient  $\lambda'_R$  is supported on squarefree integers. For squarefree  $u \leq R$ ,*

$$\lambda'_R(u) = \mu(u) \frac{u}{\varphi(u)} \sum_{\substack{b \leq R/u \\ (b,u)=1}} \frac{\mu(b)^2}{\varphi(b)}. \quad (17)$$

*In particular,*

$$|\lambda'_R(u)| \ll (\log(2R))^2. \quad (18)$$

*Proof.* Use the divisor formula

$$c_q(n) = \sum_{u|(q,n)} u\mu(q/u)$$

in (5), then write  $q = ub$  and interchange the finite sums. This gives (16) and (15). If  $u$  is not squarefree, every  $\mu(ub)$  vanishes. If  $u$  is squarefree, a nonzero summand has  $(b,u) = 1$ , and  $\mu(ub)\mu(b) = \mu(u)\mu(b)^2$ ; multiplicativity of  $\varphi$  gives (17). The standard estimates  $u/\varphi(u) \ll \log \log(3u)$  and  $\sum_{b \leq y} \mu(b)^2/\varphi(b) \ll \log(2y)$  imply (18).  $\square$

For squarefree  $u$ , define the positive weight

$$w_R(u) := \log u + \mathbf{1}_{u \leq R} \frac{u}{\varphi(u)} \sum_{\substack{b \leq R/u \\ (b,u)=1}} \frac{\mu(b)^2}{\varphi(b)}. \quad (19)$$

For  $u = 1$ , the first term is zero and the second is positive.

**Theorem 3.2** (Exact squarefree residual expansion). *For every positive integer  $n$ ,*

$$r_R(n) = - \sum_{\substack{u|n \\ u \text{ squarefree}}} \mu(u) w_R(u). \quad (20)$$

*This is a finite identity; no limiting Ramanujan expansion is used.*

*Proof.* The von Mangoldt identity

$$\Lambda(n) = - \sum_{u|n} \mu(u) \log u$$

and lemma 3.1 give

$$r_R(n) = \sum_{u|n} (-\mu(u) \log u - \lambda'_R(u)).$$

Both coefficients vanish off squarefree  $u$ , and (17) identifies the remaining coefficient with  $-\mu(u)w_R(u)$ .  $\square$

The next theorem is the main algebraic reduction. It applies row by row and hence survives every generic row-pair restriction.

**Theorem 3.3** (Primitive determinant–divisor normal form). *Let  $m_i = \ell_i d_i$  satisfy equations (7) and (8), put  $\Delta = m_1 - m_2$ , and let  $F$  be smooth and compactly supported where  $n_{m_i}(j) > 0$ . Then*

$$\begin{aligned} & \sum_{\substack{j \geq 1 \\ (j, H) = 1}} r_R(m_1 j + h) r_R(m_2 j + h) F(j/J) \\ &= \sum_{\substack{g, a, b \text{ pairwise coprime squarefree} \\ (ga, m_1 H) = (gb, m_2 H) = 1 \\ g | \Delta}} \mu(a) \mu(b) w_R(ga) w_R(gb) \sum_{\substack{j \equiv \kappa \pmod{gab} \\ (j, H) = 1}} F(j/J), \end{aligned} \quad (21)$$

where  $\kappa$  is the unique class modulo  $gab$  satisfying

$$m_1 \kappa \equiv -h \pmod{ga}, \quad m_2 \kappa \equiv -h \pmod{gb}. \quad (22)$$

The sum in (21) is locally finite. Equivalently one may impose  $ga \ll X$  and  $gb \ll X$ , with constants determined by the support of  $F$ .

*Proof.* Apply (20) to both target values. If  $u \mid m_1 j + h$ , then (10) gives  $(u, m_1 H) = 1$ ; similarly  $(v, m_2 H) = 1$ . Write

$$g = (u, v), \quad u = ga, \quad v = gb.$$

Squarefree support makes  $g, a, b$  pairwise coprime and squarefree. The two residual signs satisfy the exact cancellation

$$\mu(ga) \mu(gb) = \mu(g)^2 \mu(a) \mu(b) = \mu(a) \mu(b). \quad (23)$$

The congruences for  $j$  are

$$j \equiv -h \overline{m_1} \pmod{ga}, \quad j \equiv -h \overline{m_2} \pmod{gb}.$$

Their common modulus is  $g$ . Since  $(g, h m_1 m_2) = 1$ , compatibility is equivalent to  $\overline{m_1} \equiv \overline{m_2} \pmod{g}$ , hence to  $g \mid m_1 - m_2$ . The generalized Chinese remainder theorem then gives one class  $\kappa \pmod{gab}$ . Finally  $(gab, H) = 1$ , so that class combines independently with the  $\varphi(H)$  permitted classes modulo  $H$ . This proves (21).  $\square$

**Corollary 3.4** (Where the parity sign remains). *In the exact residual normal form, determinant content  $g$  carries no Möbius sign. All oscillation inherited from the target divisor variables is transverse:*

$$\boxed{\text{determinant content } g \text{ is unsigned, while the remaining sign is } \mu(a)\mu(b).} \quad (24)$$

Thus taking absolute values before summing  $a, b$  removes the only explicit Möbius cancellation left by the CRT reduction.

*Remark 3.5* (Why primitivity matters). The equivalence with  $g \mid \Delta$  used the individual solvability conditions and  $(g, h) = 1$ . Without the primitive mask one retains the necessary determinant condition  $g \mid h(m_1 - m_2)$ , together with the individual linear-congruence solvability conditions; it is no longer equivalent to  $g \mid \Delta$ . TPC-18 treats those nonprimitive sectors by a different endpoint-collapse mechanism; the present normal form is not silently being extended to them.

## 4 Exact primitive centering and removal of row drift

The primitive mask changes the natural finite-model mean. We record that mean exactly rather than treating it as a harmless constant. For a common period  $P$  divisible by  $H$  and by every integer at most  $R$ , write

$$\mathbb{E}_H f := \frac{1}{\#\{j \bmod P : (j, H) = 1\}} \sum_{\substack{j \bmod P \\ (j, H) = 1}} f(j). \quad (25)$$

The value is independent of the chosen common period for every  $P$ -periodic function under consideration.

**Lemma 4.1** (Exact conditional mean of the finite model). *Assume  $dH \leq R$ ,  $\ell > R$  is prime,  $d$  is squarefree, and  $(d, H) = 1$ . For  $m = \ell d$ ,*

$$\rho_{M, H, R}(m) := \mathbb{E}_H \Lambda_R(mj + h) = \frac{H}{\varphi(H)} \frac{d}{\varphi(d)}. \quad (26)$$

Equivalently, the unnormalized complete-period mean is

$$\frac{1}{P} \sum_{\substack{j \bmod P \\ (j, H) = 1}} \Lambda_R(mj + h) = \frac{d}{\varphi(d)}. \quad (27)$$

*Proof.* Average the Ramanujan expansion in (5) over allowed  $j$ . If a squarefree  $q \leq R$  has a prime factor outside  $dH$ , the corresponding local Ramanujan wave has mean zero. If  $q \mid dH$ , then  $q \leq dH \leq R$  and  $mj + h$  is coprime to  $q$  on every allowed class, so  $c_q(mj + h) = \mu(q)$ . Therefore

$$\mathbb{E}_H \Lambda_R(mj + h) = \sum_{q \mid dH} \frac{\mu(q)^2}{\varphi(q)} = \frac{dH}{\varphi(dH)} = \frac{H}{\varphi(H)} \frac{d}{\varphi(d)}.$$

Multiplication by the density  $\varphi(H)/H$  of the allowed classes gives (27).  $\square$

The calibrated prime density on those same classes is

$$\rho_{P, H}(m) := \frac{H}{\varphi(H)} \frac{m}{\varphi(m)}. \quad (28)$$

This is a local progression density, not a claim of a prime number theorem for the long modulus  $m$ .

Define the locally calibrated row signals

$$P_m(j) = \Lambda(mj + h) - \rho_{P, H}(m), \quad M_m(j) = \Lambda_R(mj + h) - \rho_{M, H, R}(m). \quad (29)$$

**Proposition 4.2** (Matched centering identity). *Under the hypotheses of lemma 4.1,*

$$r_R(mj + h) = P_m(j) - M_m(j) + \delta_H(m), \quad (30)$$

where

$$\delta_H(\ell d) = \frac{H}{\varphi(H)} \frac{d}{\varphi(d)(\ell - 1)}. \quad (31)$$

Consequently  $\delta_H(\ell d) \ll_{h,\varepsilon} X^\varepsilon/L$ .

*Proof.* Subtract (26) from (28). Since  $\ell > R \geq d$ , the prime  $\ell$  does not divide  $d$ , and

$$\frac{\ell d}{\varphi(\ell d)} - \frac{d}{\varphi(d)} = \frac{d}{\varphi(d)} \left( \frac{\ell}{\ell - 1} - 1 \right) = \frac{d}{\varphi(d)(\ell - 1)}.$$

The standard bound  $d/\varphi(d) \ll X^\varepsilon$  proves the estimate.  $\square$

**Theorem 4.3** (The drift has a power saving at the obstruction scale). *Assume  $L \geq X^{\lambda_0}$  for a fixed  $\lambda_0 > 0$ , and let a primitive row-pair correlation have the geometry and bounded weights of section 2. Replacing every  $r_R(mj + h)$  by*

$$Z_m(j) := P_m(j) - M_m(j) \quad (32)$$

*changes the correlation by*

$$O_{h,\varepsilon}(XD X^\varepsilon) + O_{h,\varepsilon}(XD/L X^\varepsilon). \quad (33)$$

*For  $\varepsilon < \lambda_0/3$ , this has a fixed power saving relative to  $XQ = XLD$ , and hence arbitrary logarithmic saving.*

*Proof.* On the support, the soft divisor bounds give  $|P_m(j)| + |M_m(j)| \ll X^\varepsilon$ , hence  $|Z_m(j)| \ll X^\varepsilon$ . By (13), the total weighted row mass is  $O(QX^\varepsilon)$ , while the same mass with one drift inserted is  $O(QX^\varepsilon/L) = O(DX^\varepsilon)$ . There are  $O(J)$  values of  $j$ . Thus all one-drift terms are

$$\ll J(Q/L)QX^\varepsilon \asymp XDX^\varepsilon,$$

and the two-drift term is

$$\ll J(Q/L)^2 X^\varepsilon \asymp (XD/L)X^\varepsilon.$$

An off-diagonal or generic row restriction can only decrease these absolute majorants.  $\square$

*Remark 4.4* (What has been centered). The function  $M_m$  has exact zero conditional mean over its finite period. The definition of  $P_m$  removes the predicted primitive progression density, but no finite-interval prime equidistribution or zero-mean statement for  $P_m$  is asserted. The hard problem is to compare the locally calibrated prime spectrum with the exactly centered model spectrum in  $Z_m = P_m - M_m$ .

## 5 The connected finite-model determinant spectrum

We now compute the complete-period covariance of the centered model  $M_m$  in (29). The result identifies exactly which modulus interactions survive centering.

**Lemma 5.1** (Centered divisor form). *Under the hypotheses of lemma 4.1, one has on the allowed classes  $(j, H) = 1$*

$$M_m(j) = \sum_{\substack{u \leq R \\ (u, mH)=1}} \lambda'_R(u) \left( \mathbf{1}_{u|mj+h} - \frac{1}{u} \right). \quad (34)$$



*Proof.* The divisor expansion (16) may be restricted to  $(u, mH) = 1$  by (10). Conditional on  $(j, H) = 1$ , a congruence  $mj \equiv -h \pmod{u}$  with  $(u, mH) = 1$  occupies density  $1/u$ . Summing those densities gives the exact mean in (26), so subtraction proves the formula.  $\square$

**Theorem 5.2** (Connected determinant covariance). *Let  $m_i = \ell_i d_i$  satisfy the primitive row hypotheses and  $d_i H \leq R$ . Put  $\Delta = m_1 - m_2$  and*

$$\kappa_R(m_1, m_2) := \mathbb{E}_H[M_{m_1}(j)M_{m_2}(j)]. \quad (35)$$

Then

$$\kappa_R(m_1, m_2) = \sum_{\substack{u, v \leq R \\ (u, m_1 H) = (v, m_2 H) = 1}} \frac{\lambda'_R(u)\lambda'_R(v)}{uv} \left( (u, v) \mathbf{1}_{(u, v) | \Delta} - 1 \right). \quad (36)$$

Every term with  $(u, v) = 1$  vanishes individually.

*Proof.* Fix  $u, v$  in (34) and set  $g = (u, v)$ . The two congruences for  $j$  are compatible exactly when  $g \mid \Delta$ , by the same primitive CRT calculation as in theorem 3.3. When compatible, their conditional joint density is

$$\frac{1}{[u, v]} = \frac{g}{uv};$$

the modulus  $H$  is independent because  $(uv, H) = 1$ . The product of the two individual densities is  $1/(uv)$ . Expanding the centered indicators gives the bracket in (36). If  $g = 1$ , that bracket is  $1 - 1 = 0$ .  $\square$

Since  $\lambda'_R$  is squarefree-supported, write

$$u = ga, \quad v = gb, \quad g, a, b \text{ pairwise coprime and squarefree.} \quad (37)$$

Then (36) becomes

$$\kappa_R(m_1, m_2) = \sum_{\substack{g, a, b \text{ pairwise coprime squarefree} \\ ga, gb \leq R \\ (ga, m_1 H) = (gb, m_2 H) = 1}} \frac{\lambda'_R(ga)\lambda'_R(gb)}{g^2 ab} \left( g \mathbf{1}_{g | \Delta} - 1 \right). \quad (38)$$

**Corollary 5.3** (Pointwise connected-kernel bound). *If  $\Delta \neq 0$ , then*

$$|\kappa_R(m_1, m_2)| \ll (\log(2R))^6 (1 + \sigma_{-1}(|\Delta|)), \quad (39)$$

where  $\sigma_{-1}(n) = \sum_{d|n} d^{-1}$ . Hence

$$|\kappa_R(m_1, m_2)| \ll (\log(2R))^6 \log \log(3|\Delta|). \quad (40)$$

No assertion in (40) is made for  $\Delta = 0$ .

*Proof.* Use (18) in (38). The contribution of the  $-1$  part is at most

$$(\log(2R))^4 \sum_{g \geq 2} \frac{1}{g^2} \left( \sum_{a \leq R/g} \frac{1}{a} \right) \left( \sum_{b \leq R/g} \frac{1}{b} \right) \ll (\log(2R))^6.$$

For the determinant part, restrict  $g$  to divisors of  $\Delta$ , replace  $g/g^2$  by  $1/g$ , and use the same two harmonic sums. This gives (39). The classical maximal-order bound for  $\sigma_{-1}$  gives (40).  $\square$

**Proposition 5.4** (Positivity before generic pruning). *For every finite primitive row family  $\{m_\alpha\}$ , the matrix*

$$[\kappa_R(m_\alpha, m_\beta)]_{\alpha, \beta} \quad (41)$$

*is positive semidefinite.*

*Proof.* For arbitrary complex  $c_\alpha$ , complete-period averaging gives

$$\sum_{\alpha, \beta} c_\alpha \overline{c_\beta} \kappa_R(m_\alpha, m_\beta) = \mathbb{E}_H \left| \sum_{\alpha} c_\alpha M_{m_\alpha}(j) \right|^2 \geq 0.$$

□

*Remark 5.5* (Generic sums are not Gram matrices). Deleting the diagonal, same-source, near-determinant, or large-gcd edges changes the quadratic form. The generic correlation retained by TPC-18 is not positive semidefinite merely because (41) is. We use positivity only for the complete row family and never transfer it to a pruned edge set.

## 6 Direct determinant frequencies and the Möbius concentration gate

Write  $e_q(x) = \exp(2\pi i x/q)$ . The connected kernel in [equation \(38\)](#) has an elementary but useful exact spectrum.

**Proposition 6.1** (Only nonzero determinant frequencies remain). *For every integer  $g \geq 1$  and  $\Delta \in \mathbb{Z}$ ,*

$$g \mathbf{1}_{g|\Delta} - 1 = \sum_{r=1}^{g-1} e_g(r\Delta). \quad (42)$$

*Consequently*

$$\kappa_R(m_1, m_2) = \sum_{\substack{g, a, b \text{ pairwise coprime squarefree} \\ ga, gb \leq R \\ (ga, m_1 H) = (gb, m_2 H) = 1}} \frac{\lambda'_R(ga) \lambda'_R(gb)}{g^2 ab} \sum_{r=1}^{g-1} e_g(r(m_1 - m_2)). \quad (43)$$

*In particular, centering removes every  $g = 1$  contribution before any estimate is made.*

*Proof.* The complete additive-character sum is  $g \mathbf{1}_{g|\Delta}$ . Removing its  $r = 0$  term proves (42); insertion in (38) proves the second assertion. □

There is a second exact decomposition that separates genuine source dispersion from a deterministic reduced-class bias. If  $(g, \ell_1 \ell_2 d_1 d_2) = 1$ , then

$$\begin{aligned} & g \mathbf{1}_{g|\ell_1 d_1 - \ell_2 d_2} - 1 \\ &= g \left( \mathbf{1}_{\ell_1 \equiv \ell_2 d_2 \overline{d_1} \pmod{g}} - \frac{1}{\varphi(g)} \right) + \left( \frac{g}{\varphi(g)} - 1 \right). \end{aligned} \quad (44)$$

Here and below a bar denotes an inverse in the displayed modulus. The first term in (44) is the discrepancy of the prime source  $\ell_1$  in one reduced class. The second term is a fixed local bias. It is not small for a fixed small  $g$ , so the model channel cannot be declared negligible from  $j$ -centering alone. A separate global estimate would have to use additional row oscillation, or else retain the other three channels in (2).

We next isolate the bare dyadic Möbius coefficient obtained by grouping only the determinant difference. The full TPC kernel also contains the  $g, a, b$  restrictions, coprimality, generic pruning, and spectral weights; those are not encoded in the following sequence. Let  $\omega_D$  be a real smooth weight supported in an interval of length at most  $D$  inside  $[D, 2D]$ , and let  $\ell_1 \neq \ell_2$  be primes greater than  $2D$ . Define

$$A_{\ell_1, \ell_2}(n) := \sum_{\ell_1 d_1 - \ell_2 d_2 = n} \mu(d_1) \mu(d_2) \omega_D(d_1) \omega_D(d_2) \quad (45)$$

and

$$M_\ell(\alpha) := \sum_d \mu(d) \omega_D(d) \exp(2\pi i - \ell d \alpha). \quad (46)$$

**Theorem 6.2** (Injective determinant geometry and exact transform). *Under the preceding hypotheses, the map*

$$(d_1, d_2) \mapsto \ell_1 d_1 - \ell_2 d_2 \quad (47)$$

*is injective on the support of  $\omega_D$ . Therefore*

$$\sum_n |A_{\ell_1, \ell_2}(n)|^2 = \left( \sum_d \mu(d)^2 \omega_D(d)^2 \right)^2. \quad (48)$$

*With the Fourier convention  $\hat{A}(\alpha) = \sum_n A(n) \exp(2\pi i - n\alpha)$ , one also has*

$$\boxed{\hat{A}_{\ell_1, \ell_2}(\alpha) = M_{\ell_1}(\alpha) \overline{M_{\ell_2}(\alpha)}}. \quad (49)$$

*Proof.* A collision in (47) gives

$$\ell_1(d_1 - d'_1) = \ell_2(d_2 - d'_2).$$

Thus  $\ell_2 \mid d_1 - d'_1$ . The support interval has length at most  $D < \ell_2$ , so  $d_1 = d'_1$ , and then  $d_2 = d'_2$ . The  $\ell^2$ -identity follows by opening the square. Finally, Fourier inversion of the defining difference gives

$$\hat{A}(\alpha) = \sum_{d_1, d_2} \mu(d_1) \mu(d_2) \omega_D(d_1) \omega_D(d_2) \exp(2\pi i - \ell_1 d_1 \alpha) \exp(2\pi i \ell_2 d_2 \alpha),$$

which is (49) because the weights are real.  $\square$

*Remark 6.3* (What  $\ell^2$  does and does not give). Suppose  $|\omega_D| \ll 1$  and, for some fixed  $c, c' > 0$ ,  $|\omega_D(d)| \geq c$  on at least  $c'D$  squarefree integers. Then (48) is  $\asymp_{c, c'} D^2$ . Thus injectivity gives an exact normalization, not a cancellation theorem. Any gain must come from where the mass of the product in (49) lies in frequency, or from cancellation against the other locally calibrated channels.

If the factors  $\mu(d_i)$  in (45) are replaced by the permitted smooth weights and phases, the resulting Möbius-free difference coefficient has the geometry covered by Pascadi's published large-sieve theorem for smooth dispersion coefficients [6, Theorem 1.7], after dyadic and signed splitting. The general theorem is governed by a representing measure for the *pair sequence*  $A_{\ell_1, \ell_2}$ , through a frequency-concentration functional [6, Theorem 5.2]. It does not act separately on the two factors  $M_{\ell_i}$ .

*Remark 6.4* (Open sharp-Möbius concentration gate). For the present coefficient, the representing measure for  $A_{\ell_1, \ell_2}$  must encode the sharp factors  $\mu(d_i)$ ; (49) identifies its Fourier density as a product of two Möbius exponential sums. The required scale parameter and rational-frequency windows depend on the preceding matched Kuznetsov/dispersion reduction, which has not yet been constructed. We therefore do not state a numerical frequency-concentration hypothesis in isolation. The open gate is to produce that matched reduction and then prove the exact Theorem 5.2 concentration bound it requires, with sufficient saving for (72). This remark is a route specification, not a conditional theorem.

*Remark 6.5* (Two missing pieces, not one). [Remark 6.4](#) is not by itself a theorem about the original residual. One must first produce a Kuznetsov or dispersion formula in which the  $PP, PM, MP, MM$  terms share the same spectral kernel and remain combined. Only then can a frequency-concentration estimate be applied. Current results for smooth dispersion coefficients, triply well-factorable weights, or separated bilinear divisor-bounded weights do not supply that matched reduction for the sharp prime–Möbius coefficient left by TPC-18 [4, 5].

## 7 CRT–Poisson duality and the long-lcm gate

The direct spectrum in [section 6](#) is organized by the determinant. Completing the orbit variable  $j$  gives a dual description in inverse phases. The factor  $H$  in the true period is kept explicit throughout.

For admissible  $u, v$ , set

$$C_{m,u}(j) := \mathbf{1}_{u|m_j+h} - \frac{1}{u} \quad (50)$$

and

$$K_{u,v}^{m_1,m_2}(j) := \mathbf{1}_{(j,H)=1} C_{m_1,u}(j) C_{m_2,v}(j). \quad (51)$$

Put  $g = (u, v)$  and

$$T = H[u, v]. \quad (52)$$

The kernel in (51) is  $T$ -periodic. With

$$\widehat{K}(r) := \frac{1}{T} \sum_{x \bmod T} K(x) e_T(-rx), \quad (53)$$

its zero coefficient is exact.

**Proposition 7.1** (Zero mode of the centered periodic kernel). *If  $(u, m_1 H) = (v, m_2 H) = 1$ , then*

$$\widehat{K}(0) = \frac{\varphi(H)}{H} \frac{1}{uv} \left( g \mathbf{1}_{g|m_1-m_2} - 1 \right). \quad (54)$$

*Thus the complete  $j$ -zero mode is the same connected determinant spectrum as in [theorem 5.2](#), multiplied only by the density of primitive classes.*

*Proof.* Conditional on  $(j, H) = 1$ , the two divisibility events have individual densities  $1/u$  and  $1/v$ . Their joint density is  $g/(uv)$  when  $g \mid m_1 - m_2$ , and zero otherwise. Multiplying the resulting conditional covariance by  $\varphi(H)/H$  proves (54).  $\square$

For  $F \in C_c^\infty(\mathbb{R})$ , use

$$\widehat{F}(\xi) = \int_{\mathbb{R}} F(x) e(-x\xi) dx. \quad (55)$$

Fourier expansion of  $K$ , followed by Poisson summation, gives the exact identity

$$\sum_{j \in \mathbb{Z}} K(j) F(j/J) = J \sum_{r \bmod T} \widehat{K}(r) \sum_{k \in \mathbb{Z}} \widehat{F}\left(J \left(k - \frac{r}{T}\right)\right). \quad (56)$$

The pair  $(r, k) = (0, 0)$  contributes  $J \widehat{F}(0) \widehat{K}(0)$ .

**Theorem 7.2** (Uniform completion of a short periodic kernel). *Fix  $\eta > 0$ , let  $\mathcal{F}$  be a uniformly bounded subset of  $C_c^\infty(\mathbb{R})$ , and take  $F \in \mathcal{F}$ . If*

$$H[u, v] \leq JX^{-\eta}, \quad (57)$$

then, for every  $A > 0$ ,

$$\sum_{j \in \mathbb{Z}} K_{u,v}^{m_1, m_2}(j) F(j/J) = J \widehat{F}(0) \frac{\varphi(H)}{H} \frac{1}{uv} \left( g \mathbf{1}_{g|m_1-m_2} - 1 \right) + O_{A, \eta, \mathcal{F}}(X^{-A}). \quad (58)$$

The error remains  $O_A(X^{-A})$  after summing any fixed polynomial number of rows and divisor variables carrying polynomially bounded weights.

*Proof.* For every term in (56) other than  $(0, 0)$ , the distance of  $k - r/T$  from zero is at least  $1/T$ . Hence its Fourier argument has size at least  $J/T \geq X^\eta$ . Repeated integration by parts gives  $\widehat{F}(\xi) \ll_N (1 + |\xi|)^{-N}$ , uniformly in the chosen family. Since  $|\widehat{K}(r)| \ll 1$ , choosing  $N$  after the fixed polynomial multiplicity proves the assertion.  $\square$

**Corollary 7.3** (A strict-margin range for the whole finite model). *Every divisor pair in the centered finite model is completable by theorem 7.2 if*

$$HR^2 \leq JX^{-\eta}. \quad (59)$$

Fix  $\eta_0 > 0$ . With  $R = X^{1/2-\delta} + O(1)$ ,  $J \asymp X/Q$ , and fixed  $h$ , suppose

$$Q \leq X^{2\delta-\eta_0}. \quad (60)$$

Then for every fixed  $0 < \eta < \eta_0$  and all sufficiently large  $X$ , condition (59) holds. Consequently, for each primitive row pair,

$$\begin{aligned} \sum_{\substack{j \in \mathbb{Z} \\ (j, H)=1}} M_{m_1}(j) M_{m_2}(j) F(j/J) \\ = J \widehat{F}(0) \frac{\varphi(H)}{H} \kappa_R(m_1, m_2) + O_A(X^{-A}), \end{aligned} \quad (61)$$

with arbitrary power accuracy after the polynomial row summation.

*Proof.* Since  $u, v \leq R$ , one has  $H[u, v] \leq Huv \leq HR^2$ . The first assertion follows from theorem 7.2 and (34); the sum of the zero modes is exactly (36). Moreover,

$$\frac{HR^2}{J} \ll_h QX^{-2\delta} \leq X^{-\eta_0+o(1)} \leq X^{-\eta}$$

for every fixed  $\eta < \eta_0$  and sufficiently large  $X$ . This proves the strict-margin statement.  $\square$

When (59) fails, the same calculation gives an exact decomposition: split the divisor pairs according to

$$H[u, v] \leq JX^{-\eta} \quad \text{or} \quad H[u, v] > JX^{-\eta}. \quad (62)$$

The first range is replaced by its connected zero mode with arbitrary power accuracy. The second range remains an explicit nonzero Poisson spectrum. This is a short/long split, not a bound for the long range. The same termwise split applies to the locally finite all-modulus residual form in theorem 3.3, but (60) only completes the finite-model divisors  $u, v \leq R$ .

We finally display the inverse phase in the compatible joint-residue term. Let  $u = ga$ ,  $v = gb$  as in (37), assume  $g \mid m_1 - m_2$ , and let  $\kappa \bmod gab$  solve (22). For each reduced class  $c \bmod H$ , CRT combines  $j \equiv c \pmod{H}$  and  $j \equiv \kappa \pmod{gab}$ . The Poisson phase of the resulting class modulo  $Hgab$  is

$$\begin{aligned} e_H(rc \overline{gab}) e_{gab}(r \overline{H} \kappa) \\ = e_H(rc \overline{gab}) e_g(-hr \overline{H} \overline{m_1} \overline{ab}) e_a(-hr \overline{H} \overline{m_1} \overline{gb}) e_b(-hr \overline{H} \overline{m_2} \overline{ga}). \end{aligned} \quad (63)$$

Every inverse in (63) is taken in the modulus of its own character, with the convention  $e_1(\cdot) = 1$  when one of  $g, a, b$  equals 1. For  $H = 1$ , the first factor disappears and  $\overline{H} = 1$ .

*Remark 7.4* (The phase belongs to one channel). The three inverse factors in (63) describe the compatible joint-indicator term. The centered product  $C_{m_1,u}C_{m_2,v}$  also contains two single-indicator terms and a constant term. Their Poisson spectra must be retained with the joint term until the four-channel matching has occurred. Applying a Kloosterman estimate only to (63) would not estimate the locally calibrated residual combination.

Classical spectral, bilinear, and trilinear Kloosterman estimates become relevant only after this phase has been produced [1–3, 11]. The present long range has three coupled squarefree moduli, the determinant condition  $g \mid m_1 - m_2$ , and sharp Möbius/source coefficients. We do not claim that this coefficient geometry satisfies an existing theorem.

## 8 A rowwise componentwise no-go theorem

The finite model has a nonzero connected spectrum even for a primitive, distinct-source row pair. The following exact example prevents a common but invalid proof strategy.

**Theorem 8.1** (An exact nonzero model witness). *Take*

$$R = 12, \quad h = 2, \quad d_1 = d_2 = 3, \quad \ell_1 = 31, \quad \ell_2 = 41. \quad (64)$$

*Then  $H = 2$ ,  $m_1 = 93$ ,  $m_2 = 123$ , and all primitive centering hypotheses hold. The exact conditional covariance is*

$$\kappa_{12}(93, 123) = \frac{433}{450}. \quad (65)$$

*If  $P = \text{lcm}(1, \dots, 12) = 27720$ , then for every integer  $T \geq 1$ ,*

$$\sum_{\substack{1 \leq j \leq TP \\ (j,2)=1}} M_{93}(j)M_{123}(j) = TP \frac{433}{900}. \quad (66)$$

*Thus the centered finite-model component need not have logarithmic, or even  $o(TP)$ , cancellation along these repeated complete periods.*

*Proof.* Insert  $R = 12$ ,  $m_1 = 93$ ,  $m_2 = 123$ , and  $\Delta = -30$  in the finite rational sum (36). Exact simplification gives  $433/450$ . The functions  $M_{93}$  and  $M_{123}$  are periodic modulo  $P$ . There are  $P/2$  odd classes, so the sum over one full period is  $(P/2)(433/450) = P(433/900)$ . Repetition proves (66). The accompanying certificate described in section 10 verifies both the divisor formula and a direct complete-period Gram calculation using exact rational arithmetic.  $\square$

**Corollary 8.2** (No rowwise componentwise substitute). *Complete-period centering alone does not force the MM channel to have zero complete-period mean for every admissible finite primitive distinct-source parameter choice. Consequently a proof of cancellation in*

$$Z_{m_1}Z_{m_2} = P_{m_1}P_{m_2} - P_{m_1}M_{m_2} - M_{m_1}P_{m_2} + M_{m_1}M_{m_2} \quad (67)$$

*cannot obtain the required saving separately for all four terms from complete-period  $j$ -cancellation alone.*

*Proof.* The MM assertion already fails by theorem 8.1. Equation (67) is the expansion of  $(P_{m_1} - M_{m_1})(P_{m_2} - M_{m_2})$ . Therefore a valid argument must either exploit additional cancellation over the row variables or combine the nonzero local model spectrum with the other channels before taking absolute values.  $\square$

*Remark 8.3* (What the witness does not prove). The witness is finite and exact. It disproves a proposed uniform rowwise componentwise lemma; it does not show that the actual residual correlation is large, that an asymptotic generic row family has a positive main term, or that determinant dispersion is false. The other three terms in (67) may cancel the model spectrum. Nor does the fixed- $R$  example rule out a rowwise asymptotic as  $R \rightarrow \infty$ , or cancellation after averaging the actual TPC row coefficients. It proves only that complete-period centering is not an algebraic rowwise annihilator. Spectral matching is one possible open route.

There is also a structural reason not to recover positivity after row-pair pruning. The full matrix  $[\kappa_R(m_\alpha, m_\beta)]$  is a Gram matrix by [proposition 5.4](#); removing diagonal, near-determinant, or large-divisor-gcd entries is a Schur mask not known to preserve positive semidefiniteness. Hence neither positivity nor the witness may be transferred automatically to the generic TPC-18 subgraph.

For orientation only, the certificate contains two deterministic sharp-window diagnostics. With the explicitly recorded finite windows and normalization, the four locally calibrated channel values are

| channel             | $(R, L, D, J) = (30, 70, 3, 80)$ | $(60, 130, 4, 100)$         |
|---------------------|----------------------------------|-----------------------------|
| $PP$                | $1.0313935 \times 10^{-2}$       | $5.6077583 \times 10^{-2}$  |
| $PM$                | $7.4448828 \times 10^{-3}$       | $5.7773558 \times 10^{-2}$  |
| $MP$                | $7.4448828 \times 10^{-3}$       | $5.7773558 \times 10^{-2}$  |
| $MM$                | $3.5400940 \times 10^{-3}$       | $5.8418970 \times 10^{-2}$  |
| $PP - PM - MP + MM$ | $-1.0357367 \times 10^{-3}$      | $-1.0505627 \times 10^{-3}$ |

These decimal data are regression tests for the implemented channel bookkeeping, not asymptotic evidence. The windows are sharp and are not the smooth weights of the theorems. In particular, the machine-readable output sets `prime_asymptotic_evidence=false`.

## 9 The high-prime local tail and the exact remaining gates

We isolate only the local Euler tail contributed by individual primes  $p > R$ . This is not the whole arithmetic tail of the sharp  $q \leq R$  Ramanujan cutoff: composite squarefree moduli  $q > R$  built entirely from primes at most  $R$  are also omitted and are not controlled by the theorem below. Let  $m_1 \neq m_2$ , put

$$A = m_1 m_2 (m_1 - m_2), \quad (68)$$

and, for each prime  $p$ , let

$$\nu_p = \#\{j \bmod p : (m_1 j + h)(m_2 j + h) \equiv 0 \pmod{p}\}. \quad (69)$$

**Theorem 9.1** (Elementary high-prime Euler tail). *Assume  $R > \max(2, |h|)$ . Then*

$$\left| \log \prod_{p > R} \frac{1 - \nu_p/p}{(1 - 1/p)^2} \right| \ll \frac{1}{R} + \frac{\log(2|A|)}{R \log R}. \quad (70)$$

*The implied constant is absolute.*

*Proof.* Since  $p > R > |h|$ , no such prime divides  $h$ . If  $p \nmid A$ , the two affine forms have two distinct roots, so  $\nu_p = 2$ , and

$$\log \frac{1 - 2/p}{(1 - 1/p)^2} = O(p^{-2}). \quad (71)$$

If  $p \mid A$ , then  $0 \leq \nu_p \leq 2$  and the same logarithm is  $O(p^{-1})$ . Hence the left side of (70) is at most

$$O\left(\sum_{p>R} p^{-2}\right) + O\left(\sum_{\substack{p>R \\ p \mid A}} p^{-1}\right).$$

The first sum is  $O(R^{-1})$ . For the second, use

$$\frac{1}{p} \leq \frac{\log p}{R \log R} \quad (p > R), \quad \sum_{p \mid A} \log p \leq \log |A|,$$

with  $\log(2|A|)$  covering  $|A| = 1$ .  $\square$

*Remark 9.2* (A local statement only). [Theorem 9.1](#) says that individual primes above  $R$  do not hide a large unrecorded local Euler factor. It says nothing about the omitted composite moduli just described, and it is not an estimate for a finite  $j$ -correlation, a long-lcm Poisson remainder, a prime-pair sum, or any of the four locally calibrated channels. In particular, it does not replace determinant dispersion.

We can now state the remaining target without ambiguity. Let  $\mathcal{G}$  be the generic row set in (11) and let  $\mathcal{C}_D(\mathcal{G})$  be (12) with the actual TPC-18 coefficients.

**Definition 9.3** (Primitive residual-dispersion target). For a prescribed  $B > 0$ , the target estimate is

$$|\mathcal{C}_D(\mathcal{G})| \ll_{B,h,\delta,\kappa} XQ(\log X)^{-B} \quad (72)$$

uniformly in the surviving dyadic parameters. This definition is a named missing estimate, not a theorem of the present paper.

The exact identities above split attempts at (72) into two auditable routes.

#### Direct determinant-frequency route.

Derive a common Kuznetsov/dispersion form for the four terms in (67), before absolute values, so that their local main spectra cancel. For the resulting pair-sequence measure, verify the precise hypotheses and concentration functional of [6, Theorem 5.2], as described in [remark 6.4](#). The explicit matched reduction and the sharp concentration estimate are both missing.

#### CRT–Poisson long-lcm route.

Apply the all-modulus residual CRT expansion termwise, or first derive an explicitly matched four-channel analogue. Evaluate every short-period zero mode and bound the complete complementary long spectrum while keeping the joint, two single, and constant terms together. [Corollary 7.3](#) supplies only the finite  $MM$  subcalculation. The missing long estimate must tolerate the coupled  $g, a, b$  moduli, the determinant constraint, and the sharp source–Möbius coefficients typified by (63).

These two matched-spectrum routes are sufficient, not logically exhaustive. A third possibility would be a separate global estimate for each channel that exploits cancellation over the full row family. [Theorem 8.1](#) says only that such an estimate cannot come from rowwise complete-period centering of the  $MM$  term.

*Remark 9.4* (Route ledger, not a closure theorem). The exact reductions identify the following sufficient future packages:

- (i) an explicitly derived matched four-channel spectral reduction, followed by verification of the precise concentration inequality required for its resulting measure, with total error  $O_B(XQ(\log X)^{-B})$ ;



- (ii) an all-modulus residual or matched four-channel Poisson decomposition, exact evaluation of all short zero modes, and a bound  $O_B(XQ(\log X)^{-B})$  for the complete long spectrum.

If a future work supplies either package with the displayed total error, then [theorems 3.3 and 4.3](#) show that the primitive block reaches (72). This is a sufficiency ledger, not a quantitatively stated conditional result: neither package is supplied here. [Theorem 9.1](#) separately records only that the local-factor ledger has no large omitted high-prime tail; it supplies no correlation estimate.

*Remark 9.5* (What closing this ledger would mean). Even a proof of (72) closes only the primitive residual block delivered by TPC-18. A twin-prime theorem would still require every preceding decomposition, endpoint, positivity, and parameter-uniformity step in the full chain. Conversely, a proof that a precisely stated concentration inequality fails in the required parameter range, or construction of a long-lcm packet that quantitatively violates the target bound, would be a rigorous obstruction result, not evidence for or against the twin-prime conjecture.

## 10 Finite dynamical interpretation, exact certificate, and conclusion

The term *spectrum* here has a literal finite-dynamical meaning. For each divisor pair, translation

$$\tau : j \mapsto j + 1 \quad \text{on } \mathbb{Z}/T\mathbb{Z}, \quad T = H[u, v], \quad (73)$$

is a finite measure-preserving dynamical system. Its Koopman eigenfunctions are  $e_T(rj)$ . The coefficients  $\widehat{K}(r)$  in (53) are therefore the exact spectral coordinates of the centered local sieve observable. The direct identity (42) diagonalizes the shared determinant factor, whereas (56) couples that finite spectrum to the long orbit window. The required four-channel cancellation is consequently a matched-spectral statement; no claim of a spectral realization of zeta zeros is involved.

### 10.1 Exact computational certificate

The standard-library Python certificate is located at

`experiments/tpc19_certificate.py`.

Run it from the paper directory with

```
python experiments/tpc19_certificate.py
python -0 experiments/tpc19_certificate.py
```

Both commands regenerate `experiments/tpc19_certificate.json`. The proof layer uses integers, `fractions.Fraction`, and formal prime-log indeterminates. In particular, it checks:

- (i) the Ramanujan definition of  $\Lambda_R$  against the divisor form in 540 exact cases, including squarefree support and (17);
- (ii) the primitive masked means, the covariance formula, and termwise cancellation of every  $g = 1$  contribution;
- (iii) equality of a four-row direct complete-period Gram matrix and the CRT matrix, all 15 principal minors, and 625 integer-vector quadratic forms;
- (iv) the positive witness 433/450 and a separate negative covariance regression  $-259/900$ ;

- (v) the full residual identity (20) for  $n \leq 360$ , and direct/divisor/CRT formal-polynomial reassembly of a residual pair, including determinant support and common- $g$  sign cancellation.

The direct periodic side loops over the original  $q$ -truncated Ramanujan model, whereas the comparison side assembles the  $u$ -divisor and CRT forms. These are independently looped reorganizations, not wholly independent arithmetic libraries: the Ramanujan-sum routine itself uses the standard divisor formula for  $c_q(n)$ .

At the version archived with this paper, the generated JSON has SHA-256

B141ADC152C38CDABE46528BD872AA4622AFD7A8E9D1E8E719D283247AC05744,

and the script has SHA-256

60154632DB45B1DB023C8CF5BF6D4084481EEF9154851CD2121567C2DA96F61F.

The optional prime-data diagnostics are isolated from every exact check and explicitly serialize `prime_asymptotic_evidence=false`.

## 10.2 Conclusion

TPC-19 advances the TPC chain by replacing an opaque variable- determinant residual correlation with exact, reusable normal forms. The actual residual has a positive-size squarefree divisor expansion with transverse Möbius signs. Primitive CRT identifies the shared divisor with the row determinant. Exact centering removes a small row drift. The complete-period  $j$ -zero mode of the finite model is supported on nonzero determinant frequencies, and CRT–Poisson duality turns its long-period part into coupled inverse phases.

The negative theorem is equally informative: one local model channel can have a nonzero complete-period mean, so rowwise orbit cancellation is unavailable. A future proof can seek extra oscillation over rows, or preserve matched cancellation and then pass either the sharp Möbius frequency-concentration gate or the joint long-lcm Kloosterman gate. These are narrower targets than the TPC-18 residual statement, but they remain open. The paper therefore delivers a rigorous step and an auditable route map, not a twin-prime claim or a parity-barrier breakthrough.

## References

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